SADC HUMANITARIAN AND EMERGENCY OPERATIONS CENTRE

TRAINING MANUAL

ICT & Emergency Telecommunications Protocol

Based on SOP 3.1 (Communication Infrastructure Setup)
and SOP 3.2 (Support of Communications Network) v3.0 Enhanced Edition

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SADC ICT Protocol 2026 | FEMA NIFOG | Sendai Framework | Tampere Convention

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1. Introduction and Purpose

This Training Manual is a companion document to SOP 3.1 (Communication Infrastructure Setup) and SOP 3.2 (Support of Communications Network), Version 3.0 Enhanced Edition. It provides SHOC ICT personnel, ERT-deployed ICT Specialists, and NDMA telecommunications counterparts with practical, phase-by-phase guidance on emergency telecommunications operations across the full UNDAC mission lifecycle.

The manual is structured around the six UNDAC mission lifecycle phases: **Be Aware (Phase 0), Alert (Phase 1), Standby (Phase 2), Dispatch (Phase 3), On-Mission (Phase 4), and Mission-End (Phase 5).** For each phase, the manual identifies the key tasks, the applicable SOP sections, the Annex toolkit items to use, and the performance targets to achieve.

Trainees should use this manual alongside the full SOP text and the Annexes Compendium (Document Control: SADC-SHOC-VOL3-ANNEXES-2026). The SOPs provide the authoritative procedural requirements; this manual translates those requirements into a concise operational learning guide.

2. Terms and Abbreviations

The following terms are essential to this training manual. A full list is provided in the SOP, Section 18.

Abbreviation	Full Form
AAR	After-Action Review
BGAN	Broadband Global Area Network
CAP	Common Alerting Protocol
EIC	Emergency Items Catalogue (IFRC)
ERT	Emergency Response Team
ETK	Emergency Telecommunications Kit
EW	Early Warning
GIS/DCM	Geographic Information System / Disaster Connectivity Map
ICT	Information and Communications Technology
NDMA	National Disaster Management Authority
NETP	National Emergency Telecommunications Plan
NETWG	National Emergency Telecommunications Working Group
NOC	Network Operations Centre
NOC-lite	Lightweight field-deployable NOC monitoring dashboard
OSSOC	On-Site SHOC Operations Coordination Centre
PMI	Performance Monitoring Indicator
PoA	Plan of Action
SHOC	SADC Humanitarian and Emergency Operations Centre
SitRep	Situation Report
SOC	Security Operations Centre
VSAT	Very Small Aperture Terminal

VPN	Virtual Private Network

3. How to Use This Manual

This manual follows the six-phase UNDAC mission lifecycle established in SOP 3.1 v3.0. Each phase section is structured identically to provide a consistent learning experience:

- Phase Purpose: Explains what the phase achieves and why it matters.
- Key Tasks: Summarises the critical actions you must perform.
- Annex Toolkit: Identifies which Annexes from the Compendium to use and when.
- Quick Reference Table: Provides a condensed task-output-timeline summary.
- Trainer's Notes: Highlights common errors, drill scenarios, and assessment cues.

3.1 Training Approach

Training delivery should combine classroom instruction with practical exercises. Each phase should be trained using a realistic scenario (e.g. Tropical Cyclone, Flood, Earthquake) with participants completing the relevant Annex templates under simulated time pressure. Quarterly Readiness Drills (SOP Phase 0, Section 6.3.4) serve as both operational readiness checks and competency assessments.

3.2 Document Cross-Reference Conventions

Throughout this manual, references to SOP sections appear as ***"SOP §X.Y"*** and references to Annexes as ***"Annex 3.1-X"*** or ***"Annex 3.2-X"***. All Annexes are published in the companion Volume 3 Annexes Compendium.

4. Phase 0: Be Aware — Continuous Telecom Readiness

SOP Reference: SOP §6 (Sections 6.1–6.3.5)

4.1 Phase Purpose

Phase 0 operates continuously, 24/7, and is not triggered by any crisis event. It ensures that SHOC's emergency telecommunications capability maintains deployment-ready status at all times. The UNDAC Handbook (p. 66) is clear: the receipt of an alert is not the time to start thinking about what to bring or who to call.

4.2 Key Tasks

1. Continuous ETK Status Monitoring — Maintain automated polling of battery (min. 80%), firmware, antennas, subscriptions, and SIMs. Resolve amber alerts within 24 hours; red alerts within 8 hours. (SOP §6.3.1)
2. Hazard-to-Telecom Profile Mapping — Maintain pre-configured deployment profiles for each SADC hazard type (cyclone, flood, drought, earthquake, epidemic, conflict, industrial, compound). Link to INFORM Sub-national Risk Index. (SOP §6.3.2)
3. Frequency Pre-Clearance Cache — Maintain standing clearances, regulatory contacts for all 16 Member States, Tampere Convention packages, and customs document packs. Verify monthly. (SOP §6.3.3)

4. Quarterly Readiness Drills — Simulate complete activation-to-OSSOC-deployment sequence, testing ETK mobilisation, triple-redundancy failover, NOC-lite setup, and SHOC-to-OSSOC handoff. (SOP §6.3.4)
5. Member State Readiness Matrix — Record NETP domestication, EW4All status, alerting authorities, regulator readiness, and customs facilitation readiness per Member State. Classify as Mature, Transitional, or Assisted. (SOP §6.2.1)

4.3 Annex Toolkit for Phase 0

Annex	Title	Use in Phase 0
3.1-A	Emergency Telecom Deployment Checklist	Pre-verify readiness status quarterly
3.1-D	Asset Tracking & Serial Register	Maintain current inventory of all ETKs
3.1-G	Quarterly Readiness Drill Report	Document drill results and corrective actions
3.2-F	Redundancy Drills & Failover Testing	Record monthly and quarterly drill outcomes
3.1-S	Member State Readiness Matrix	Track NETP status and focal-point registers
3.1-Y	Training & Sustainability Matrix	Track certifications and budget readiness

4.4 Trainer's Notes

Common errors: Treating Phase 0 as administrative rather than operational; allowing ETK battery levels to drop below 80%; failing to update regulatory contacts after personnel changes in Member States. During training exercises, set a scenario where an ETK returns a red alert and require the trainee to resolve it within the 8-hour window using the correct escalation pathway.

5. Phase 1: Alert — Parallel Activation

SOP Reference: SOP §7 (Sections 7.1–7.3.5)

5.1 Phase Purpose

Phase 1 replaces the previous sequential model with a parallel activation architecture: ICT mobilisation commences simultaneously with ERT roster alerts, not after them. This eliminates the sequential delay where telecommunications mobilisation began only after the Activation Order.

5.2 Key Tasks

Task	Action Required	Target	Annex
Parallel ICT Alert	ICT Module auto-triggers alert to all telecom personnel via SMS, email, and ERT Portal	≤ 1 hour	—
Disaster-Specific ETK Config	Hazard-to-Telecom Mapper auto-selects optimal ETK configuration for hazard type and terrain	≤ 4 hours	—
ICT Specialist Confirmation	ICT specialists confirm availability via ERT Portal	≤ 2 hours	—

NDMA In-Country Assessment	Remote assessment of host country telecom infrastructure, regulatory environment, OSSOC site	≤ 4 hours	3.1-H
Frequency Pre-Clearance	Activate relevant pre-clearances; submit disaster-specific frequency requests	≤ 6 hours	3.2-A
Formal Request Routing	Open SHOC support-request record; file Member State request through SADC Secretariat	≤ 2 hours	3.1-T
NETWG/Regulator Notification	Notify NETP focal point, regulator, and NETWG; validate roaming and public warning requirements	≤ 6 hours	—

5.3 Trainer's Notes

Critical learning point: Trainees must understand that three operational streams activate in parallel upon alert: (a) Situation Room issues the formal alert, (b) ICT team activates pre-staging, and (c) ERT Coordinator initiates roster alerts. Simulate a scenario where the NDMA assessment (Annex 3.1-H) reveals no Tampere Convention ratification in the host country and require the trainee to identify the customs and regulatory implications.

6. Phase 2: Standby — Team and Kit Preparation

SOP Reference: SOP §8 (Sections 8.1–8.2.4)

6.1 Phase Purpose

The Standby phase finalises team composition, assembles the OSSOC Telecom Go-Kit, and develops the telecommunications contribution to the ERT Plan of Action. The UNDAC Handbook explicitly includes telecommunications equipment and solutions as a key logistics consideration within the Plan of Action.

6.2 Key Tasks and Annexes

Task	Annex to Complete	Target
Prepare Telecom PoA Section (connectivity plan, ETK manifest, frequencies, cybersecurity)	3.1-I: Telecom PoA Section Template	≤ 12 hours
Assemble OSSOC Telecom Go-Kit per disaster-specific config; pack for air transport	— (use 3.1-D for asset tracking)	≤ 18 hours
Verify all equipment against IFRC EIC specifications	3.1-K: IFRC EIC Verification Checklist	≤ 24 hours
Confirm frequency pre-clearances activated during Alert phase	3.2-A: Frequency Allocations	≤ 24 hours
Pre-configure VPN profiles and activate cloud mirror	—	≤ 24 hours
Decide surge escalation tier (SHOC-only, host-state, ITU Harare, or partner)	3.1-X: Surge Escalation Decision Tree	≤ 18 hours
Prepare customs documentation pack for dispatch	3.1-U: Customs/Border Facilitation Pack	≤ 24 hours

6.3 OSSOC Telecom Go-Kit Contents

The standard Go-Kit includes: one primary VSAT or Starlink terminal with auto-pointing antenna; one BGAN terminal as backup; two VHF handheld radios; one HF radio; one ruggedised laptop with NOC-lite software (SolarWinds/Zabbix lightweight instance); pre-configured VPN profiles; UPS and portable solar charging system; complete cable kit and tool bag; and the OSSOC Connectivity Readiness Checklist (Annex 3.1-J).

Equipment staging follows a four-tier escalation model: Tier 1 (SHOC internal assets only), Tier 2 (SHOC plus host-state assets), Tier 3 (SHOC plus ITU Harare augmentation), Tier 4 (SHOC plus partner-supported surge assets).

7. Phase 3: Dispatch — Deployment

SOP Reference: SOP §9 (Sections 9.1–9.2.4)

7.1 Phase Purpose

The Dispatch phase transitions from preparation to deployment. It introduces Component 6a (a dedicated Telecommunications Briefing) to the Pre-Deployment Briefing and formalises the SHOC-to-OSSOC Telecom Handoff Protocol. The deployment target is OSSOC telecommunications connectivity within 8 hours of ERT arrival.

7.2 Key Tasks and Annexes

Task	Annex	Target
Deliver Component 6a Telecom Briefing (connectivity plan, frequencies, cybersecurity, handoff, fallback)	3.1-L	During Pre-Deployment Briefing
Execute SHOC-to-OSSOC Handoff Protocol; ICT Specialist signs Accountability Transfer Form	3.1-M	Pre-departure
Record dispatch in ICT Deployment Register; activate GPS tracking; upload to SHOC Portal	3.1-D	≤ 2 hours pre-departure
Create encrypted Signal group for ERT team; configure SHOC-OSSOC voice/data channels	—	≤ 1 hour pre-departure
Transmit customs pack and border facilitation documents to handlers and receiving authorities	3.1-U	Pre-departure

7.3 Trainer's Notes

Assessment focus: The Accountability Transfer Form (Annex 3.1-M) is a formal custody document. Trainees must understand that once signed, all ETK and Go-Kit items are their personal accountability. Simulate a scenario where a Go-Kit item is found to have a firmware mismatch at the point of handoff and require the trainee to identify the correct resolution sequence before signing.

8. Phase 4: On-Mission — Operations

SOP Reference: SOP §10 (Sections 10.1–10.3.2)

8.1 Phase Purpose

Phase 4 encompasses ERT arrival, OSSOC establishment, sustained operations, and handover planning initiation. The critical performance target is full OSSOC telecommunications connectivity within 8 hours of arrival, supported by NOC-lite monitoring, independent failover protocols, and a continuous early warning feed-back loop.

8.2 Deployment Sequence (Critical Timeline)

Window	Task	Key Actions	Annex
0–2 hours	Rapid Site Assessment	Evaluate line-of-sight, RF interference, power, physical security at OSSOC location	—
2–6 hours	Primary Connectivity	Install VSAT/Starlink; establish VPN to SHOC NOC; commission backup paths; test interoperability	3.1-N
6–8 hours	NOC-Lite & Go-Live	Configure NOC-lite dashboard; set auto-failover triggers; issue Connectivity Readiness Certificate	3.1-J
8+ hours	Sustained Operations	Early Warning feed-back loop; daily SitReps; SOC-NOC monitoring; public warning integration	3.1-E

8.3 Daily Operational Cycle

Once the OSSOC is declared operational (Annex 3.1-J signed), the ICT Specialist enters a daily cycle: complete the Daily Telecom SitRep (Annex 3.1-E) by 18:00 daily; verify early warning feed-back loop functionality; coordinate with NDMA on public warning activation where required (Annex 3.1-V); produce GIS/DCM telecom decision-support maps (Annex 3.1-W); and maintain the SOC-NOC integrated monitoring log. The SitRep is uploaded to the SHOC Portal and feeds the Situation Room Dashboard.

8.4 Trainer's Notes

Practical exercise: Set up a tabletop exercise where primary VSAT connectivity fails at hour 4. Require trainees to execute failover to BGAN backup within 30 seconds (auto-failover target), document the event in Annex 3.1-N, report via SitRep (Annex 3.1-E), and notify SHOC NOC. Test whether trainees correctly prioritise P1 (Command & Control) traffic per Annex 3.2-B during degraded operations.

9. Phase 5: Mission-End — Transition and Lessons Learned

SOP Reference: SOP §11 (Sections 11.1–11.2.5)

9.1 Phase Purpose

Mission-End encompasses transition planning, operational handover to NDMA, demobilisation, and the After-Action Review contribution. Handover and exit planning should begin from the last third of the deployment, not at the end.

9.2 Key Tasks and Annexes

Timeline	Task	Annex
14 days before end	Develop Telecom Transition Plan: assets for NDMA transfer vs. SHOC return; NDMA capacity assessment; training plan	3.1-O
7–3 days before end	Train NDMA staff on retained systems; transfer configs, licences, vendor contacts; joint connectivity verification	3.1-P
Final 3 days	4-step Asset Management Protocol: reconcile serial numbers, GPS locations; package returns; decommission OSSOC	3.1-Q
7–21 days post-demob	Prepare Telecom Lessons Learned Report; present at ERT AAR; enter findings into PMIS	3.1-R
21–30 days post-demob	Feed AAR findings into Phase 0: hazard profile revisions, ETK pre-positioning, drill scenario updates	—
Before final exit	Structured closeout with NETWG, NETP focal point, regulator, and alerting authority	—

9.3 Trainer's Notes

The Phase 0 Feedback Loop is the critical learning outcome: every mission must improve the next one. Trainees should demonstrate they can trace a field lesson (e.g. antenna alignment difficulty in high-wind terrain) back through the Lessons Learned Report (Annex 3.1-R) into a specific Hazard-to-Telecom Profile revision and a revised Quarterly Drill scenario.

10. Support of Communications Network (SOP 3.2)

SOP Reference: SOP §12 (Sections 12.1–12.2, Steps 1–22)

SOP 3.2 governs the continuous maintenance, resilience, and support of SHOC's telecommunications network. It operates across all mission lifecycle phases. The Enhanced Edition integrates SOP 3.2 with the six-phase lifecycle established in SOP 3.1.

10.1 Triple-Redundancy Architecture

Trainees must understand the three-layer redundancy model that underpins all SHOC communications:

Layer	Primary	Backup	Tertiary
Layer 1: Satellite	VSAT C-Band	VSAT Ku-Band	Starlink LEO
Layer 2: Terrestrial	Fibre Optic	LTE/5G failover	MPLS VPN inter-site
Layer 3: Radio	HF long-range	VHF tactical	TETRA/DMR interoperable

Target: 99.9% uptime with automated failover within 30 seconds and zero data loss (RPO=0).

10.2 Integrated SOC-NOC Model

Cybersecurity is not a standalone function; it is integrated into network operations through the SOC-NOC coordination model. When the SOC detects a threat (DDoS, intrusion, malware), the NOC automatically initiates failover to an uncompromised redundancy layer while the SOC implements containment. Trainees must be able to execute this integrated response.

10.3 SOP 3.2 Annex Toolkit

Annex	Title	Training Focus
3.2-A	Frequency Allocations & Radio Licenses	Regulatory process and pre-clearance model
3.2-B	Traffic Prioritisation & QoS Management	P1–P5 traffic classes and failover priority
3.2-C	Inter-Agency Coordination Agreements	MoU register and RACI matrix
3.2-D	Public Warning & Community Comms	CAP workflow, multilingual alerts, accessibility
3.2-E	Cybersecurity & Data Protection	Encryption standards, access control, incident response
3.2-F	Redundancy Drills & Failover Testing	Monthly and quarterly drill scheduling
3.2-G	Standard Documentation Templates	Template index and SLA KPIs
3.2-H	Scalability & Public Network Integration	Scalability thresholds and failover sequences
3.2-I	Post-Incident Review & Lessons Learned	Root cause analysis and improvement actions

11. Annex Toolkit Quick Reference

The following table maps all SOP 3.1 Annexes to their applicable mission phase, enabling rapid identification of the correct toolkit item during operations.

Annex	Title	Phase(s)
3.1-A	Emergency Telecom Deployment Checklist	0–3
3.1-B	Network Topology Diagram	4
3.1-C	Connectivity Verification Template	4
3.1-D	Asset Tracking & Serial Register	0–5
3.1-E	Daily Telecom SitRep Template	4
3.1-F	End-of-Mission Report & AAR	5
3.1-G	Quarterly Readiness Drill Report	0
3.1-H	In-Country Telecom Assessment Report	1
3.1-I	Telecom PoA Section Template	2
3.1-J	OSSOC Connectivity Readiness Certificate	2/4
3.1-K	IFRC EIC Verification Checklist	2
3.1-L	Pre-Deployment Telecom Briefing Template	3
3.1-M	Telecom Accountability Transfer Form	3
3.1-N	OSSOC Connectivity Deployment Log	4
3.1-O	Telecom Transition Plan Template	5
3.1-P	Operational Handover Certificate	5
3.1-Q	Telecom Demobilisation Record	5
3.1-R	Telecom Lessons Learned Report	5

12. Performance Monitoring Indicators

Trainees must know and be able to measure the following PMIs, which are used to assess both operational performance and training competency.

Indicator	Target	Frequency
OSSOC Connectivity Speed	≤ 8 hours from ERT arrival	Per operation
Network Uptime	≥ 99.9%	Continuous
Auto-Failover Speed	≤ 30 seconds	Per failover event
Data Integrity (RPO)	0 loss events	Continuous
ETK Readiness Rate	≥ 95%	Monthly
EW Feed-Back Loop Latency	≤ 5 minutes	Per event
Drill Completion Rate	100%	Quarterly
Formal Request Routing Compliance	100%	Per activation
Public Warning Activation Readiness	100%	Per activation
Customs Facilitation Success Rate	≥ 95%	Per deployment
Training & Certification Coverage	≥ 90%	Quarterly

13. Risk Awareness Summary

Trainees should be aware of the top operational risks identified in the SOP Enhanced Risk Analysis Matrix (SOP Section 13). The following are the highest-residual-risk items that directly affect telecommunications operations:

Risk	Description	Key Mitigation
Infrastructure Failure	Hardware or antenna malfunction during deployment	Triple-redundancy; daily diagnostics; SHOC NOC remote support
OSSOC Connectivity Failure	Unable to establish comms within 8 hours	Dedicated Go-Kit; pre-configured VPN; ICT Specialist in every ERT
Public Warning Approval Failure	CAP/CB/LB-SMS messages delayed or not authorised	Controlled alerting authority register; fallback broadcaster relay
Customs/Border Delay	Telecom kits delayed at entry/exit	Customs facilitation pack; serialised manifests; border escalation contacts
Cyber Intrusion	Malware attack on VPN or cloud nodes	Integrated SOC-NOC monitoring; 2FA; automated threat-based failover
Regulatory Interop. Failure	No roaming or emergency SIM during response	Regulator escalation ladder; pre-cleared templates; NETWG consultation

14. Assessment and Competency Verification

Competency in SHOC Emergency Telecommunications operations is assessed through a combination of practical exercises and knowledge verification aligned to the SOP phase structure.

14.1 Competency Areas

Competency	Assessment Method	Minimum Standard
ETK readiness monitoring and resolution	Practical: resolve simulated amber/red ETK alerts	Amber ≤ 24h; Red ≤ 8h
Hazard-to-Telecom profile application	Scenario: select correct profile for assigned hazard	Correct profile with justification
NDMA In-Country Assessment	Complete Annex 3.1-H under time pressure	≤ 4 hours; all sections complete
OSSOC Go-Kit assembly and EIC verification	Practical: assemble Go-Kit and complete Annex 3.1-K	100% EIC compliance confirmed
SHOC-to-OSSOC Handoff execution	Role-play: execute handoff and sign Annex 3.1-M	All items reconciled; form signed
OSSOC deployment (0–8 hour sequence)	Field/tabletop exercise: achieve connectivity	≤ 8 hours; Annex 3.1-J issued
Daily SitRep completion	Complete Annex 3.1-E from operational data	All sections complete; on time
Failover execution	Simulate primary path failure; execute failover	≤ 30 seconds; correct P1 prioritisation
Transition planning and NDMA handover	Complete Annexes 3.1-O and 3.1-P	All assets reconciled; certificate signed
Lessons Learned and Phase 0 feedback	Complete Annex 3.1-R; trace to profile revision	Actionable findings entered into PMIS

Certification requirement: ICT personnel must demonstrate competency across all ten areas before being listed as deployment-ready on the ERT roster. Quarterly Readiness Drills (Annex 3.1-G) provide the primary assessment vehicle. Results are recorded in the Training, Certification and Sustainability Matrix (Annex 3.1-Y) and reviewed by the ICT Manager.

— END OF TRAINING MANUAL —